

Hardware and Sensor Interaction

GPIO Pins are the physical plugs on the chip. An example is connecting a wire to a light bulb so you can turn it on and off.

Communication Protocols are the languages used to talk to smart parts. An example is the chip asking a digital thermometer for the exact room temperature.

Pulse Width Modulation is flipping power on and off super fast.

Interrupts are immediate panic buttons. An example is an airbag sensor stopping everything else to deploy the moment it detects a crash.

Connectivity and Networking

Wi-Fi Provisioning is connecting the device to the internet. An example is typing your home password into a new smart plug so it can get online.

Bluetooth handles short-range local connections. An example is your phone finding a new smartwatch when they are sitting right next to each other.

MQTT Protocol is a quick text messaging system for machines. An example is a water sensor texting the main hub the word wet so the hub knows to shut a valve.

REST APIs are how the chip asks the internet for favours. An example is a smart clock checking a weather website to see if it needs to display a rain icon.

User and Cloud Interfaces

Web Servers are tiny websites living inside the chip.

IoT Dashboards are apps that organise your data. An example is opening Apple Home or Google Home to look at a clean graph of your monthly power usage.

Over-The-Air Updates are wireless software fixes. An example is your smart TV downloading a security patch overnight without you plugging a cable into it.

Deep Sleep Modes are power naps for electronics. An example is a forest fire sensor sleeping for an hour, waking up for one second to check for smoke, and immediately passing out again to save the battery.

